Classification or Regression?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach two valid questions about the same evidence.

Supply complete paired values, preserve units/alignment limits and assess a fresh threshold/quantity comparison.

Keep one habit: Ask what a target number means and whether the metric answers the user's actual question.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Identify a task from the meaning of its target

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Use two 45-minute sessions to distinguish tasks by target meaning, not by **input** types or whether values contain decimals. Connect to Week 2 by keeping **input**s available at dispatch and outcomes separate.

Supplied duration estimates let students inspect category decisions and numerical errors without suggesting a model was trained. The **threshold** is fixed before evaluation. All examples are fictional.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

No. It is known later and is the outcome to predict, not an available dispatch-time **input**.

Printed lesson and workbook

Number line marked at 10 minutes and cards for no and yes

Optional Python 3 editor; no packages needed

Blank actual estimate category **absolute error** table

Opening retrieval

Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

Look for: Separate **prediction input**s from observed targets

Reference example, not the only acceptable wording: Candidate features are pages and queue_at_checkout.

The target is returned_late and the tracking field is loan_id. days_to_return is known only after return and directly defines the target, so it must not enter checkout **prediction**s.

End goal

By the end, I can explain **classification** or **regression**? using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Features, Labels, Targets

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

System A predicts **class** codes 0=oak,1=maple. System B predicts leaf area in cm². Name the task types and explain why code 1 is not 'one more species' than code 0.

Concept to explain

Classification predicts a category; **regression** predicts a numerical quantity. The target—not the **input** format—determines the task type.

Term	Meaning
classification	predict a category
regression	predict a number
class	one allowed category
continuous target	number that can vary across a range

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach target meaning before model type or metric

> Keep duration, height and encoded-category examples separately labelled.

Read dispatch questions, target types and timing

At robot dispatch, we can ask two questions about the same fictional trip. Will it take more than 10 minutes? Or how many minutes will it take? The first asks for a category. The second asks for an amount with units.

The target question determines the task, even if both models would use the same dispatch-time **input**s.

Classification predicts a category from a defined set. **Regression** estimates a numerical quantity for which differences have meaning.

A category may be coded with numbers: 0 and 1 here are names for no and yes, not travel times. Conversely, a duration recorded as a whole number of minutes can still be a **regression** target.

Decimal points do not decide the task.

Keep Week 2's **prediction** time: route length and the saved queue may exist at dispatch, while the observed duration is available later for **training** or evaluation.

Do not feed that observed duration into an earlier dispatch **prediction**.

Question	Target meaning	Output check
Will the trip exceed 10 minutes?	No/yes category; 0 and 1 are category codes	Compare category with the defined observed category
How many minutes will it take?	Numerical duration with meaningful differences	Absolute discrepancy in minutes; average error with case count
Can a duration estimate be thresholded?	Yes: this example uses estimate>10 to form a category	That does not turn a matching category into an exact duration estimate
Data pairing	Each actual/estimate pair must be the same trip	Equal lengths alone do not establish matching identities

Trace supplied duration estimates and paired errors

The following **prediction**s are supplied for practice; no model is fitted here. Each estimate was written before revealing its matching observed duration.

We compare duration error with a category decision made from that estimate. Predict slow when the estimated duration is greater than 10 minutes; exactly 10 is not slow.

A **class**ifier could also predict categories directly, but this example uses a **threshold** on estimates.

zip pairs the lists in order.

They must have equal lengths and refer to the same trips; otherwise this loop is not a valid evaluation. abs gives the distance from zero, so both a one-minute overestimate and a one-minute underestimate have **absolute error** 1.

The **output** is 2 3 followed by [1, 1, 2]. The mean **absolute error** is (1 + 1 + 2) / 3 = 4/3 minutes, about 1.33 minutes.

The 14-minute trip has the largest duration error in our example, yet its slow category is correct. The 10-minute trip has only a one-minute error but the category is wrong. These checks answer different questions.

Report 2 correct out of 3 category decisions separately from a mean **absolute error** of about 1.33 minutes. Do not compare a fraction directly with minutes.

Read the complete height pairs

Distinguish predicting growth height from predicting a growth category.

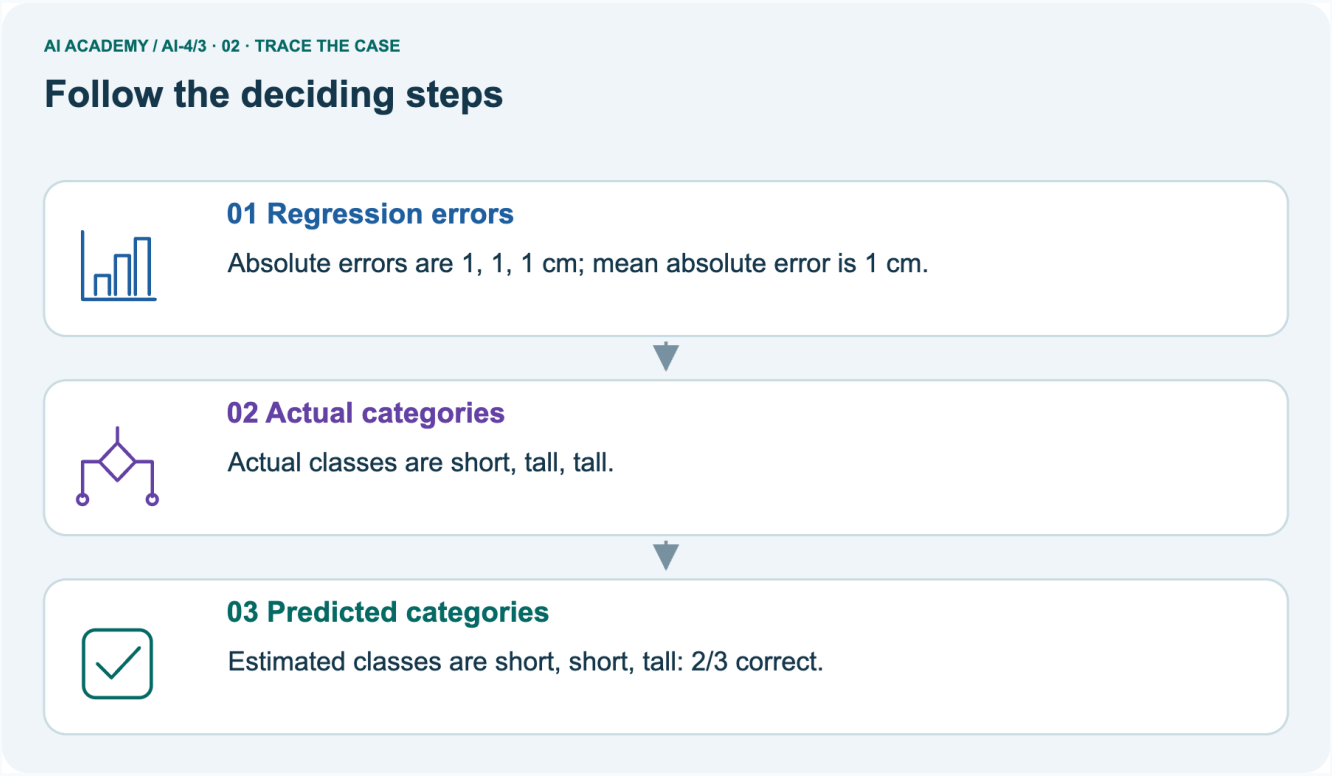
Actual heights are [4, 7, 9] cm; estimates are [5, 6, 10] cm. Tall means strictly greater than 6 cm.

All **input**s and outcomes in this worked example are invented **class**room data; no real model run is claimed.

Keep units and scenarios separate. Dispatch durations actual 8,10,14 and estimated 9,11,12 give errors 1,1,2 minutes and 2/3 categories correct at>10. Pictured heights actual 4,7,9 cm and estimates 5,6,10 cm give errors 1,1,1 cm. Strict >6 cm gives 2/3 matching categories; changing only to >7 cm gives 3/3. Height estimates and MAE 1 cm do not change.

These are measured-height targets, not growth increments; no initial-height differences are supplied. Changing the category definition is not evidence of a better numerical estimate.

Trace numerical and categorical outcomes



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/3

Actual and estimated heights (paired order)



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Regression errors	Absolute errors are 1, 1, 1 cm; mean absolute error is 1 cm.
2. Actual categories	Actual classes are short, tall, tall.
3. Predicted categories	Estimated classes are short, short, tall: 2/3 correct.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Estimated **class**es are short, short, tall: 2/3 correct. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict changing only the category boundary

Change: Change the category boundary to strictly greater than 7 cm.

Prediction: Actual and estimated categories both become short, short, tall: 3/3 correct.

Reason: The numerical estimates did not improve; the decision definition changed.

State metric/use consequences and diagram limits

Regression error and category accuracy measure different targets; neither alone covers every use consequence.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Additional task-type review

Task A predicts healthy/spotted; B predicts next week's height in cm.

Choose task and **output** types.

A is **classification** with categorical **output**; B is **regression** with numeric **output** and units. Numeric **class** codes do not make **classification regression**.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Choose the task type for six scenarios and define the **output** space precisely.

MODEL IDEA **Classification** predicts a category; **regression** predicts a numerical quantity. The target—not the **input** format—determines the task type.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Predict plant species from a photo: target is a category, so **classification**. Predict plant height next week: target is centimeters, so **regression**.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Predict plant species from a photo: target is a category, so **classification**.

STEP 2 Predict plant height next week: target is centimeters, so **regression**.

STEP 3 Predict low/medium/high growth: categories, even though ordered.

STEP 4 State allowed **class**es or numerical unit before selecting a model and metric.

CHECK Verify the denominator, unit, split, **class** definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Exact explanation to model

Explain the added worked example

Key terms

Term	Meaning
Classification	Predicting a category from a defined set
Regression	Estimating a numerical quantity with meaningful differences
Threshold	A boundary used to turn a value into a decision
Absolute error	The nonnegative distance between an estimate and an observation

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

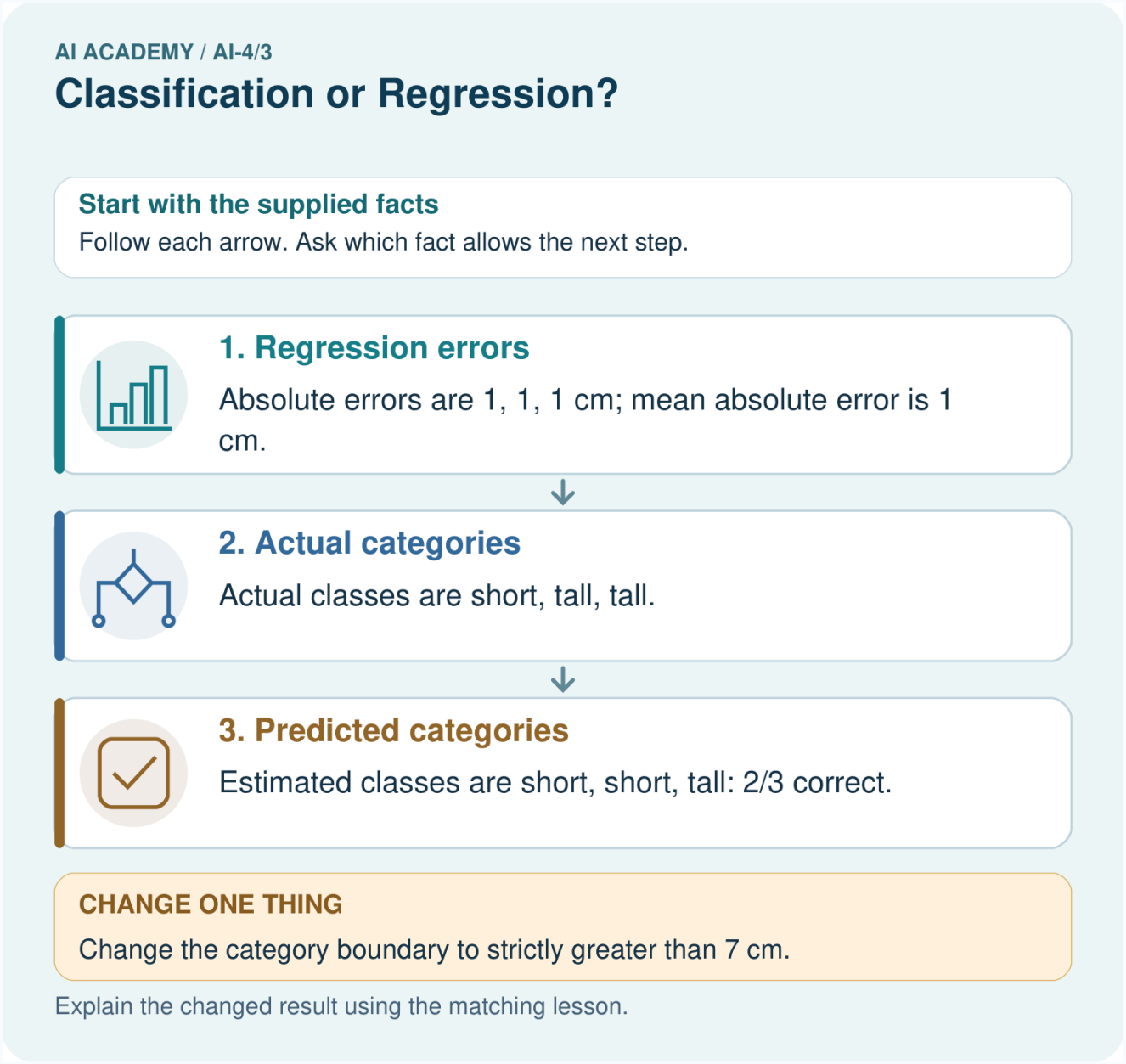
> **A trip actually takes 10 minutes and its estimate is 10.5 minutes. At the greater-than-10 boundary, give the actual category, predicted category and absolute duration error. Is the category decision correct?**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Classification or Regression?

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: For actual 14 minutes and estimate **12 minutes**, calculate **absolute error** and both slow categories using greater than 10. Explain why a correct category does not mean an exact duration.

Teacher check: **Absolute error** is 2 minutes. Both actual and predicted categories are 1, so the category matches even though the duration estimate is 2 minutes too low.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Classification** and **regression**. Proposed explanation: "Numbers always mean **regression**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Category codes remain categories. Ask what a difference between the numbers would mean in the task.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the **class**.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Did **regression** MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

No. MAE remains 1 cm. Category agreement rose because the **class** boundary changed, not because **prediction**s became numerically closer.

Changed-case check: Actual and estimated categories both become short, short, tall: 3/3 correct. The numerical estimates did not improve; the decision definition changed.

Changed-case reasoning: Actual and estimated categories both become short, short, tall: 3/3 correct. The numerical estimates did not improve; the decision definition changed.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Task A predicts healthy/spotted; B predicts next week's height in cm.

Write the question to explain: Choose task and **output** types.

Reveal after **prediction**s: A is **classification** with categorical **output**; B is **regression** with numeric **output** and units. Numeric **class** codes do not make **classification regression**.

Use the earlier transfer question as additional practice. Reserve Compare the new print-job evidence independently for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: actual category 0, predicted category 1, absolute duration error 0.5 minutes; the category is wrong.

Exit: specify **regression** for duration in minutes and use a numerical error check such as mean **absolute error** in minutes. Category agreement does not report how close the duration estimates are to observed times.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the Week 2 rule that later outcomes cannot be dispatch inputs. Ask the prerequisite question and keep the prediction moment visible.
7 to 18 minutes	Compare the three target questions. Ask why numeric category codes remain categories and why whole-minute durations remain quantities.
18 to 32 minutes	Trace the three supplied estimates. Explain abs and zip, including equal lengths and matching record order. Complete W1 and W2.
32 to 45 minutes	Use the number line for the guided 10 versus 10.5 case. Separate category agreement from numerical closeness.
Session 2 start to 8 minutes	Retrieve task meaning and units. Ask for a numerical category code and a whole-number quantity as counterexamples to the decimal shortcut.
8 to 20 minutes	Complete W3 and discuss information lost by binning. Distinguish directly predicting a class from thresholding a quantity estimate.
20 to 35 minutes	Collect W4 independently, requiring category lists, denominator, absolute errors and the mean with units. Record hints or other support.

Stage	Teacher action
35 to 45 minutes	Use W5 and the exit to match outputs to a user need. Offer the fresh retry after support where necessary.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Clinic: 1 healthy/spotted classification; 2 height quantity regression; 3 route codes classification; 4 box-count quantity regression under declared discrepancy meaning; 5 vague target needs definition; 6 ordered class classification. Height probes MAE 1 cm unchanged while category 2/3→3/3 under new cutoff; reject incomplete pairing or mismatched units.

Independent actual [4,5,9], estimated [6,5,8], slow>5: categories [0,0,1]/[1,0,1], 2/3 correct; errors [2,0,1], MAE 1 minute. No training or universal result.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. Task-Type Clinic: **class**ify six problem statements, rewrite vague targets, and match each to a useful metric.

3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

Numbers always mean **regression**: Category codes remain categories. Ask what a difference between the numbers would mean in the task.

Regression requires decimal **observation**s: A quantity measured or rounded to whole minutes still has meaningful numerical distances. Target meaning decides the task.

A correct category means a precise estimate: The 14 versus 12 minute example has a correct category but a two-minute error. Conversely, a small error near the boundary can change the category.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve Compare the new print-job evidence independently for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR A photograph always means **classification** and a table always means **regression**.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: For three fictional print jobs, actual completion times are 4, 5 and 9 minutes; estimates are 6, 5 and 8 minutes in matching order. Slow means more than 5 minutes.

Write actual and predicted category lists, count correct categories, calculate each **absolute error** and the mean error in minutes. State what each result measures.

Actual categories are [0, 0, 1]; predicted categories are [1, 0, 1], so 2 of 3 match. **Absolute error**s are [2, 0, 1] minutes; their mean is 1 minute.

The fraction measures **threshold**-category agreement on these jobs, while mean **absolute error** measures average duration discrepancy.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Classify the target meaning

At dispatch, identify the task for predicting route type coded 1 or 2, duration in whole minutes, and whether duration exceeds 10 minutes. Explain your choices.

Route type is **classification** because 1 and 2 encode categories. Whole-minute duration is **regression** because numerical differences measure time. Exceeding 10 minutes is binary **classification** because the target is yes or no.

W2 Trace both checks

For actual 14 minutes and estimate **12 minutes**, calculate **absolute error** and both slow categories using greater than 10. Explain why a correct category does not mean an exact duration.

Absolute error is 2 minutes. Both actual and predicted categories are 1, so the category matches even though the duration estimate is 2 minutes too low.

W3 Repair a category average

Indoor is coded 1 and outdoor is coded 2. Someone averages **prediction**s and reports route type 1.5 as an accurate halfway category. Explain the problem and name an appropriate check.

The codes label categories, so 1.5 is not a defined route type. Predict one of the defined categories and compare it with the observed category. A numerical midpoint of arbitrary codes does not establish correctness.

W5 Specify a useful output

A user needs the arrival duration in minutes, but a report gives only slow or not slow. What task should be specified, and what information was lost by the category **output**?

Specify **regression** for duration in minutes and check numerical error, such as mean **absolute error** in minutes.

The category omits the estimated duration and how far it is from the boundary; it cannot answer the requested quantity.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

A is **classification** with categorical **output**; B is **regression** with numeric **output** and units. Numeric **class** codes do not make **classification regression**.

Additional transfer practice question

Task A predicts whether a plant is fern or grass. Task B predicts its height in centimetres. Which is **classification** and which is **regression**?

Reasoned practice answer

A is **classification** because it selects a category; B is **regression** because it predicts a numerical quantity.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a **class**room checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification **threshold**.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Any task involving plant measurements is **regression**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

A is **classification**; B is **regression**. The codes name categories and do not establish arithmetic differences between species.

Area is a numeric target with meaningful units and distances; choose evaluation appropriate to each **output**.

Explanations for the additional workbook examples

W1 Classify the target meaning

W2 Trace both checks

W3 Repair a category average

W4 Compare new evidence independently

For three fictional print jobs, actual completion times are 4, 5 and 9 minutes; estimates are 6, 5 and 8 minutes in matching order. Slow means more than 5 minutes.

Write actual and predicted category lists, count correct categories, calculate each **absolute error** and the mean error in minutes. State what each result measures.

Actual categories are [0, 0, 1]; predicted categories are [1, 0, 1], so 2 of 3 match. **Absolute error**s are [2, 0, 1] minutes; their mean is 1 minute.

The fraction measures **threshold**-category agreement on these jobs, while mean **absolute error** measures average duration discrepancy.

W5 Specify a useful output

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Confirm whether Plant **Class**ifier predicts a health category or a numerical severity score; document why.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant **Class**ifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive **class**. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Use a number line to mark actual and estimated times, then color which side of the boundary each lies on. Supply a blank table without answers. Extension: recode no and yes as 7 and 9.

The **classification** task and matching count stay the same, while distances between those codes are not time errors.

Require W4 actual categories [0,0,1], predicted [1,0,1], 2 of 3 correct, **absolute error**s [2,0,1] and mean 1 minute. The learner must explain both meanings.

Fresh retry: actual times 3,6,8 and estimates 4,5,10; slow means greater than 6. Actual and predicted categories are both [0,0,1], so 3 of 3 match. **Absolute error**s [1,1,2] have mean 4/3 minutes, about 1.33.

Ask why perfect category agreement does not mean zero duration error.

Delayed return

Suggested return points in this level: ai-4/4, ai-4/6, ai-4/9. Use actual **class**room dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 4: Python and Table Readiness. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 4 inspects datasets and documents their origin, units and limitations. Carry forward the target definition, **prediction** moment and measurement units.

Teacher reference: https://scikit-learn.org/1.0/tutorial/basic/tutorial.html (**classification** and **regression** definitions; reviewed 2026-09-26).

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case
</details>

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Identify a task from the meaning of its target

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before **class**; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

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>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before **class**, solve the supported practice: For actual 14 minutes and estimate **12 minutes**, calculate **absolute error** and both slow categories using greater than 10.

Explain why a correct category does not mean an exact duration.

Reasoned practice answer: **Absolute error** is 2 minutes. Both actual and predicted categories are 1, so the category matches even though the duration estimate is 2 minutes too low.

Check every learner before discussion: Use the worked case for **Classification** and **regression**. Proposed explanation: “Numbers always mean **regression**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Category codes remain categories. Ask what a difference between the numbers would mean in the task.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change the category boundary to strictly greater than 7 cm. Require a **prediction** and a reason before comparing. Expected result: Actual and estimated categories both become short, short, tall: 3/3 correct.

Reason: The numerical estimates did not improve; the decision definition changed.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

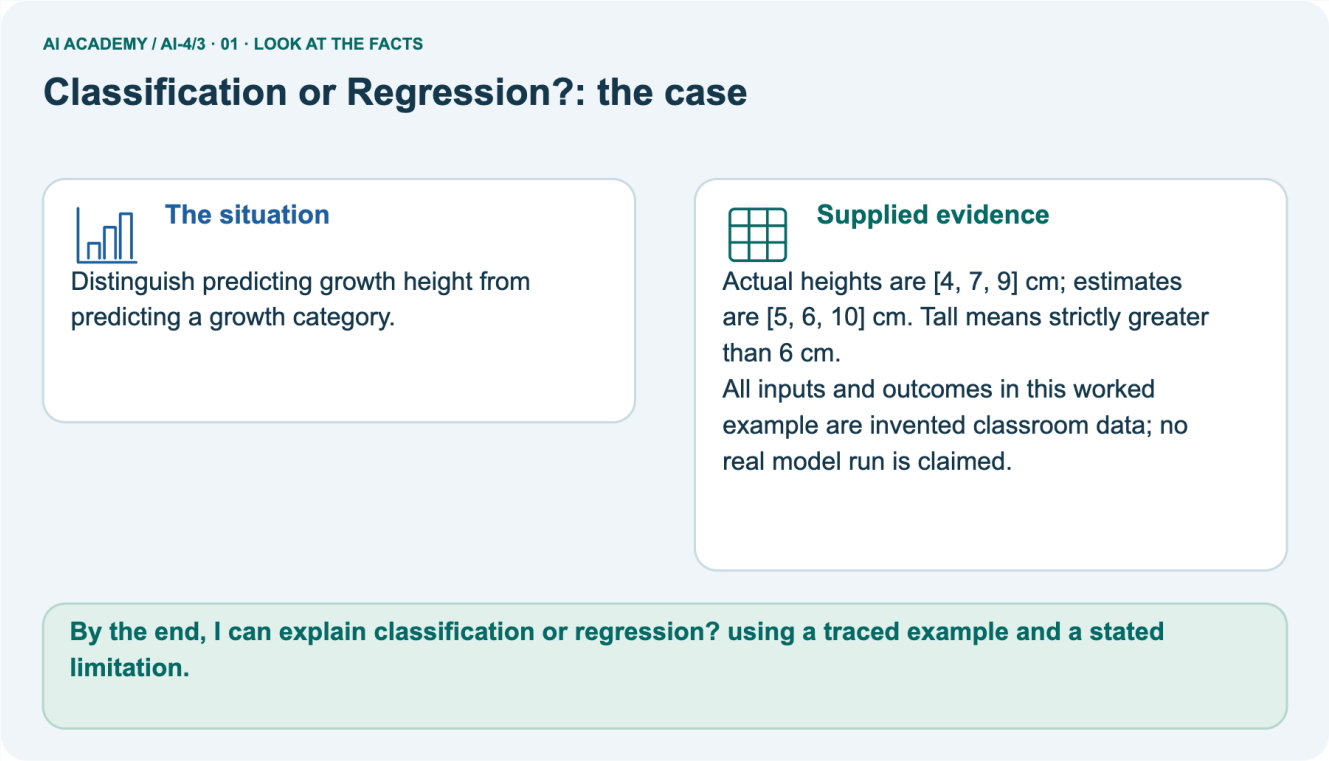
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Classification or Regression?: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

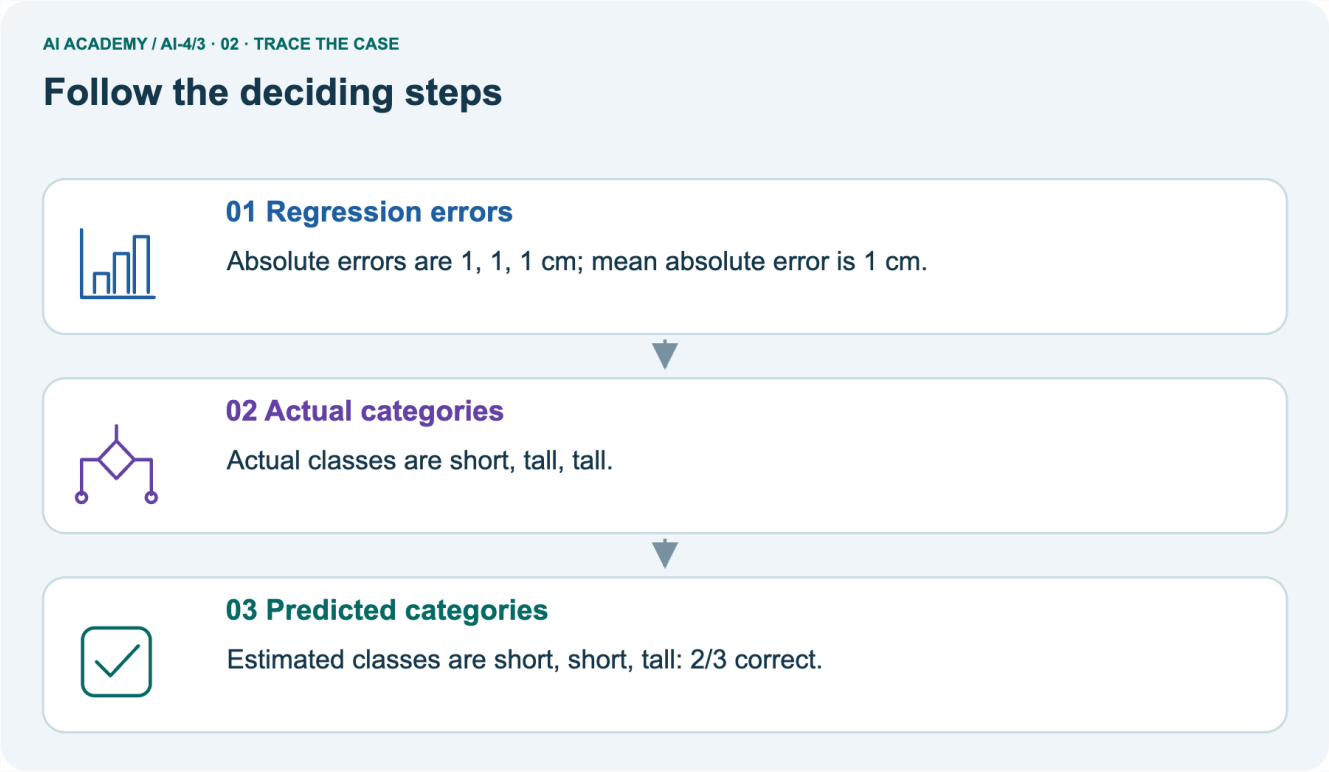
Distinguish predicting growth height from predicting a growth category. Supplied facts: Actual heights are [4, 7, 9] cm; estimates are [5, 6, 10] cm. Tall means strictly greater than 6 cm.

All **input**s and outcomes in this worked example are invented **class**room data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Regression errors: **Absolute error**s are 1, 1, 1 cm; mean **absolute error** is 1 cm. → Actual categories: Actual **class**es are short, tall, tall. → Predicted categories: Estimated **class**es are short, short, tall: 2/3 correct.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Actual and estimated heights (paired order)



Follow the labelled picture with the matching explanation. Open the vector diagram

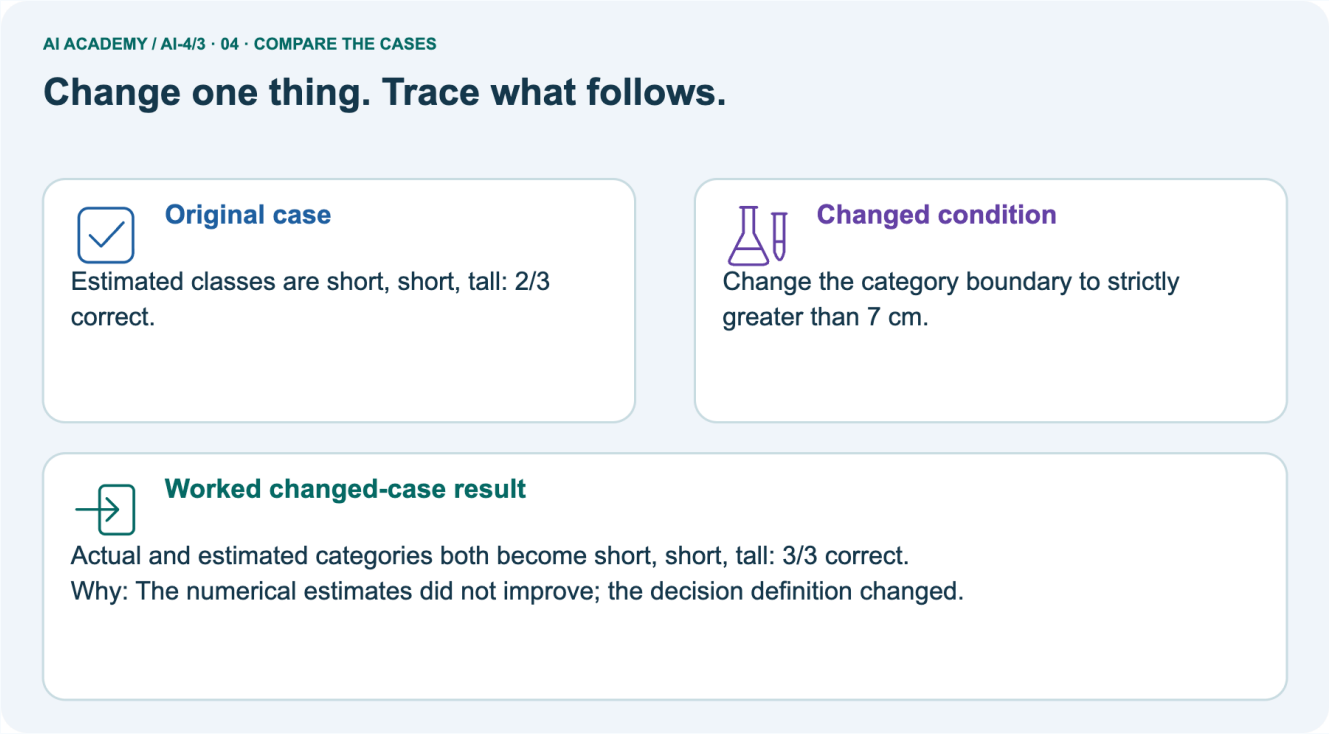
Actual and estimated heights (paired order). Units: cm. actual1: 4. estimate1: 5. actual2: 7. estimate2: 6. actual3: 9. estimate3: 10.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real **class** or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Estimated **class**es are short, short, tall: 2/3 correct. Changed condition: Change the category boundary to strictly greater than 7 cm.

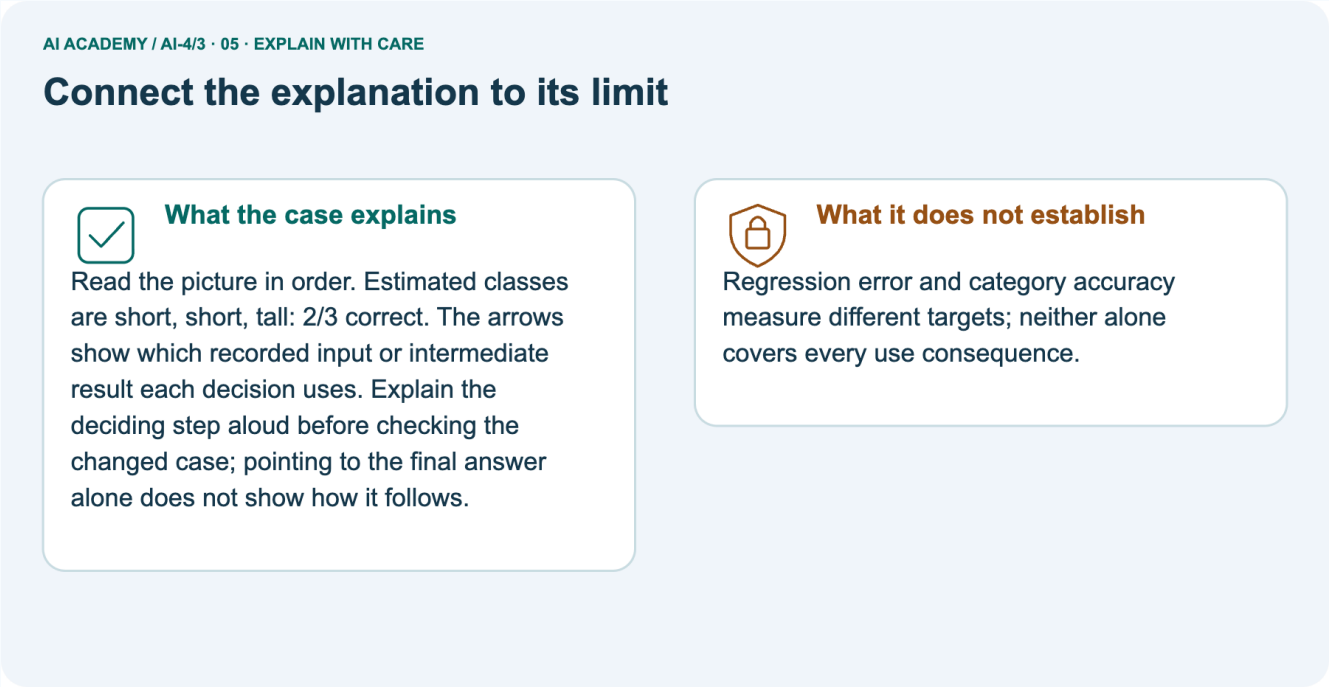
Expected result: Actual and estimated categories both become short, short, tall: 3/3 correct. Reason: The numerical estimates did not improve; the decision definition changed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The numerical estimates did not improve; the decision definition changed.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Estimated **class**es are short, short, tall: 2/3 correct. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

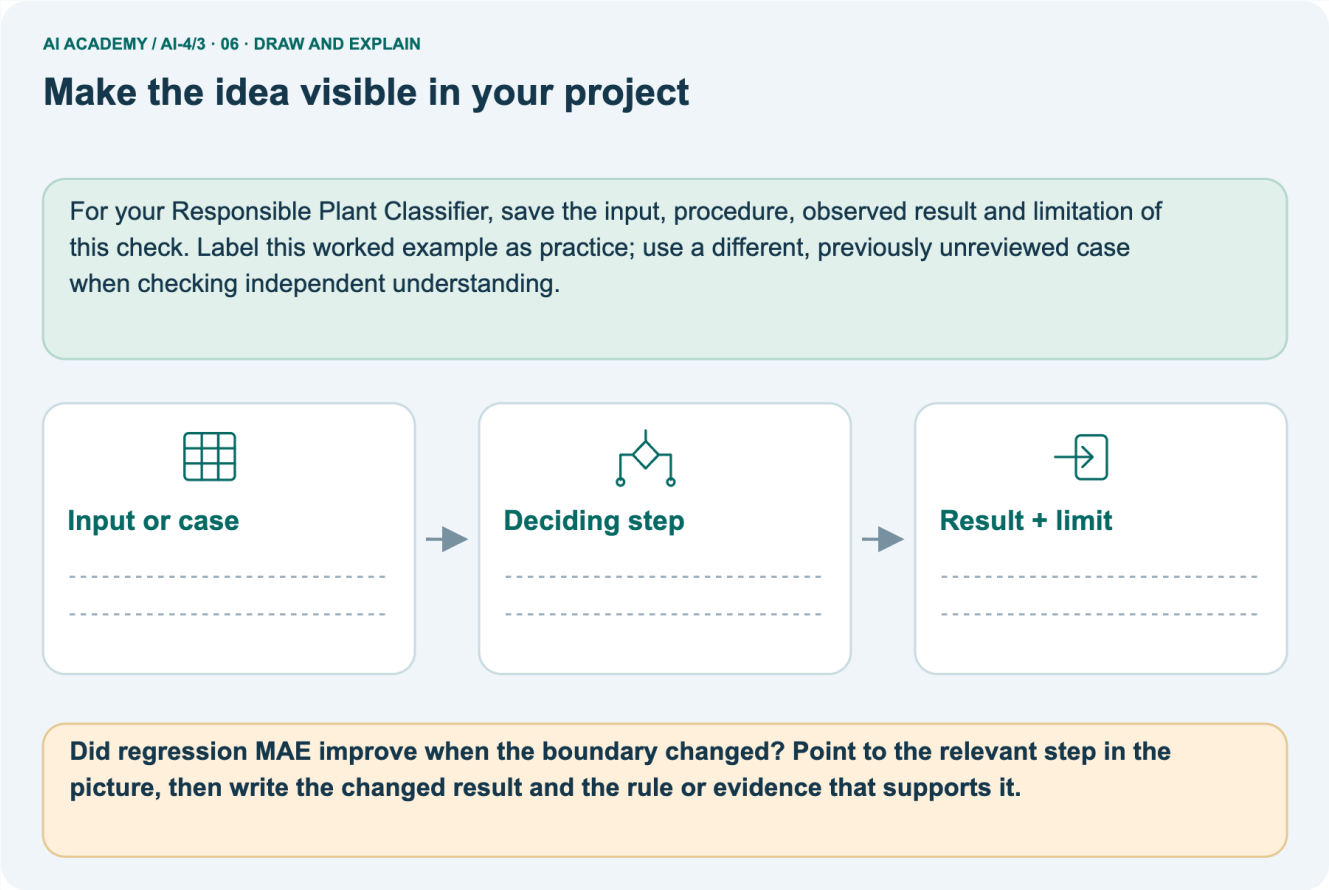
Limit: **Regression** error and category accuracy measure different targets; neither alone covers every use consequence.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Regression** error and category accuracy measure different targets; neither alone covers every use consequence.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant **Class**ifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Did **regression** MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. MAE remains 1 cm. Category agreement rose because the **class** boundary changed, not because **prediction**s became numerically closer.

Changed-case check: Actual and estimated categories both become short, short, tall: 3/3 correct. The numerical estimates did not improve; the decision definition changed.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Classification or Regression?: the case

02. Follow the deciding steps

03. Actual and estimated heights (paired order)

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.